

# Customer Churn & Retention Analysis

## Power BI Dashboard – Project Insights

### 1. Project Objective

The objective of this project is to analyze customer churn and retention patterns using the Telco Customer Churn dataset and present the findings through an interactive Power BI dashboard.

### 2. Dataset Overview

The dataset contains 7,043 customer records and includes information such as customer tenure, contract type, payment method, internet service, monthly charges, total charges, and churn status.

### 3. Key Performance Indicators

| Metric             | Value  |
|--------------------|--------|
| Total Customers    | 7,043  |
| Churned Customers  | 1,869  |
| Churn Rate         | 26.54% |
| Retained Customers | 5,174  |
| Retention Rate     | 73.46% |

### 4. Key Insights

**Contract Type:** Churn differs substantially across contract types. Month-to-month customers have an observed churn rate of approximately 42.71%, while one-year contract customers have approximately 11.27% and two-year contract customers have approximately 2.83%.

**Customer Tenure:** Churn rates vary across customer tenure groups. The dashboard groups customers into 0–12, 13–24, 25–36, 37–48, 49–60, and 61–72 month ranges to make the differences easier to compare.

**Payment Method:** The observed churn rate differs across payment methods. The dashboard provides a direct comparison of these customer groups.

**Internet Service:** Churn rates also differ across internet service categories. The dashboard compares the observed churn rate for each available service category.

### 5. Dashboard Features

The dashboard contains five KPI cards: Total Customers, Churned Customers, Churn Rate, Retained Customers, and Retention Rate. It also contains four analytical charts covering contract type, customer tenure, payment method, and internet service. Dropdown filters are provided for Contract, Internet Service, and Payment Method.

### 6. Conclusion

The analysis shows that customer churn is not uniform across customer groups. Contract type, customer tenure, payment method, and internet service category show differences in observed churn rates. These findings can support further investigation of customer retention patterns.

Tools Used: Power BI | DAX | Excel